

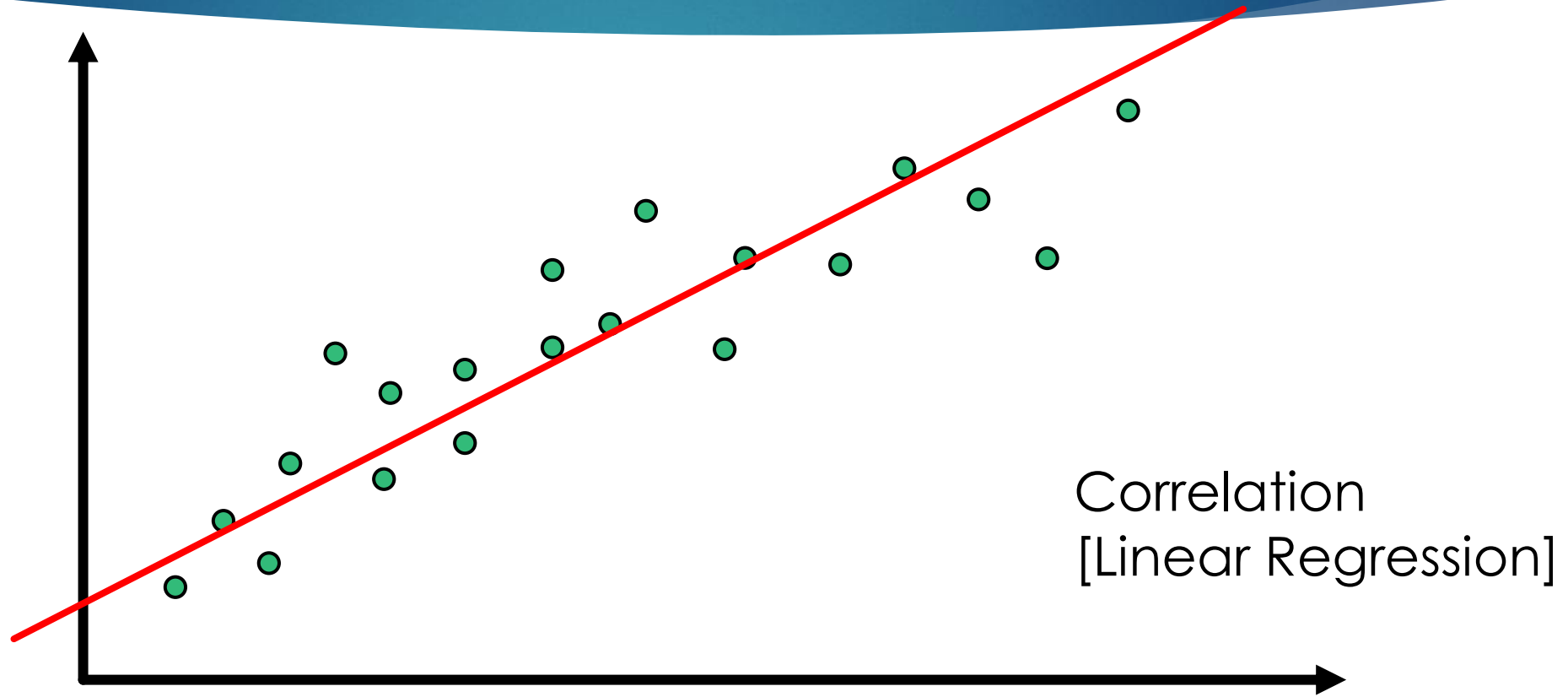


The Foundation of Correlation

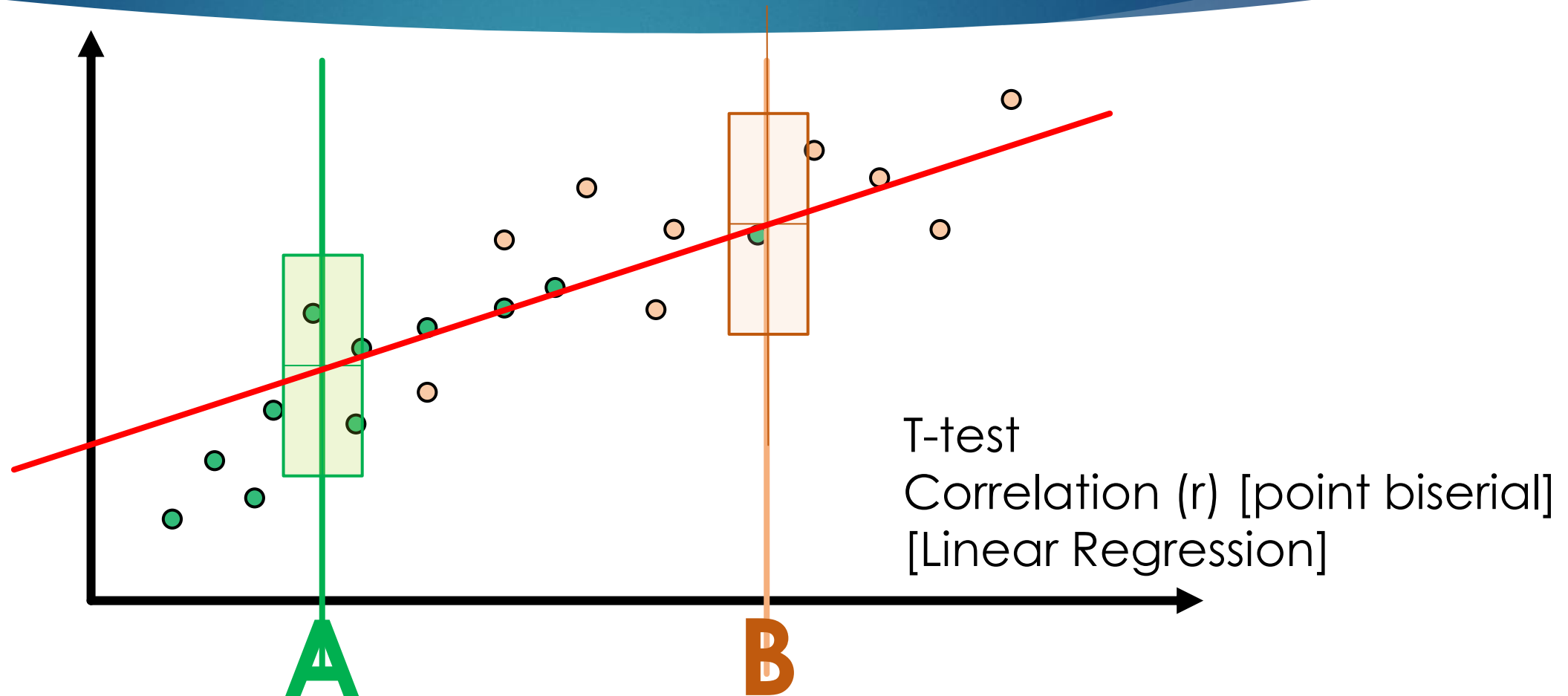
OR HOW TO UNDERSTAND THE UNDERPINNINGS OF MOST/NEARLY ALL STAT MODELS

MELINDA HIGGINS, PHD, RESEARCH PROFESSOR

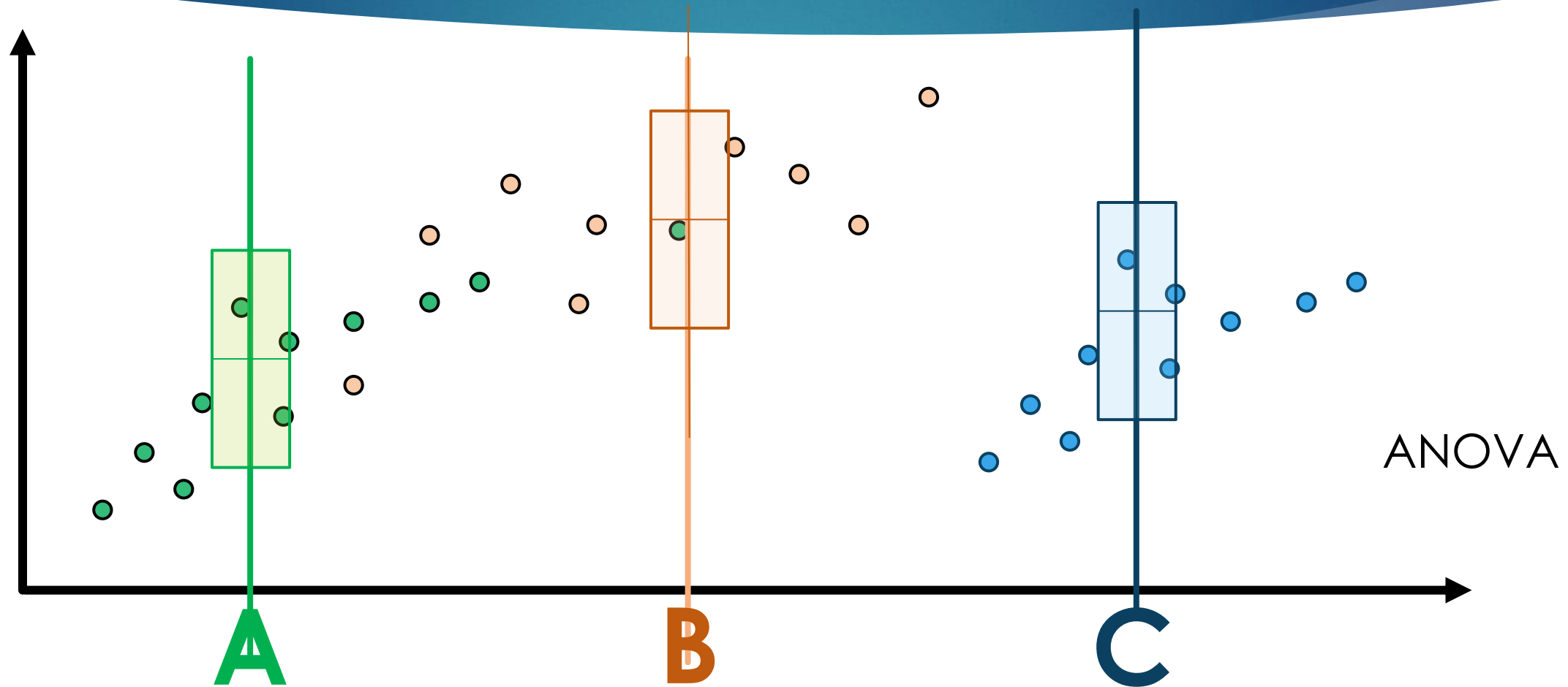
2 continuous variables



Y: Continuous Outcome
X: Categorical – 2 groups



Y: Continuous Outcome
X: Categorical – 3+ groups

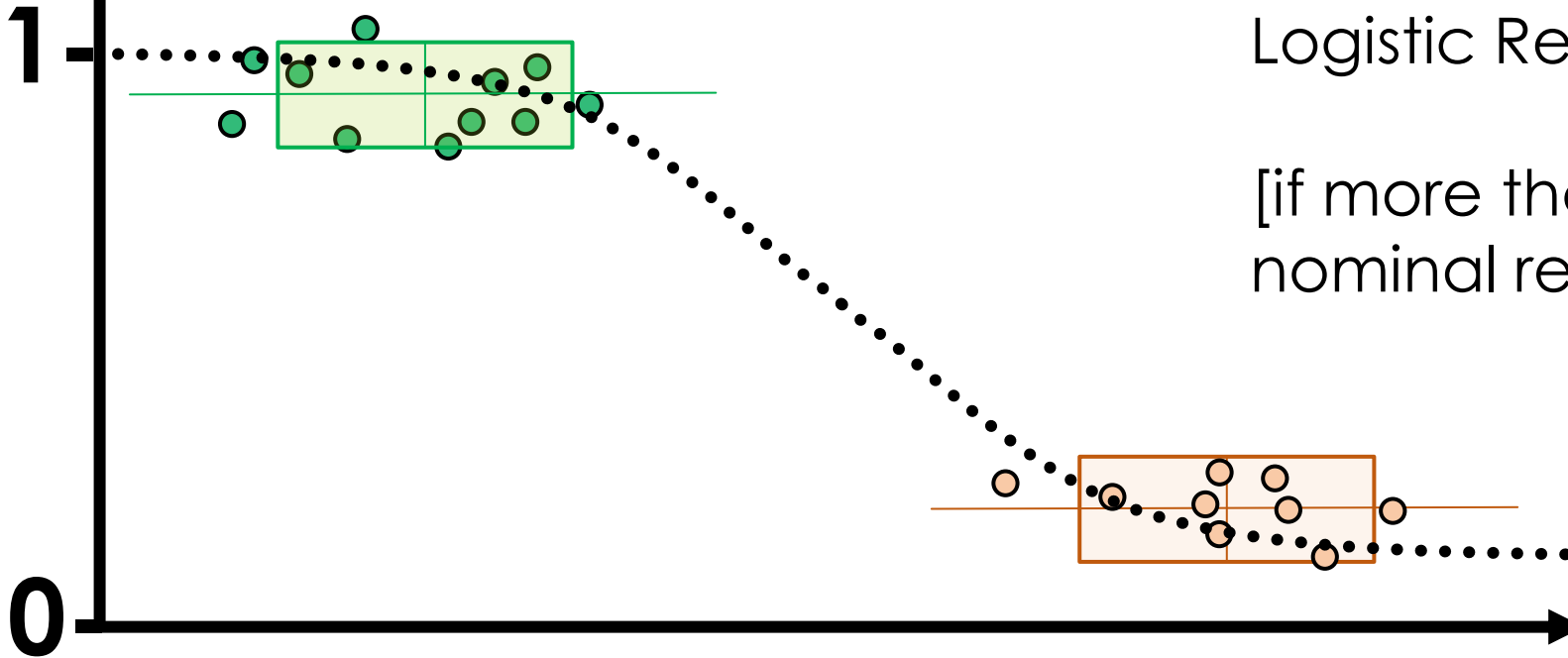


Y: Categorical – 2 groups
X: Continuous

$P(Y=YES)$

YES

NO



Logistic Regression (binary)

[if more than 2 – ordinal or
nominal regression]

Y: Categorical – 2 groups

X: Categorical – 2 (or more) groups

$P(Y=YES)$

YES

1

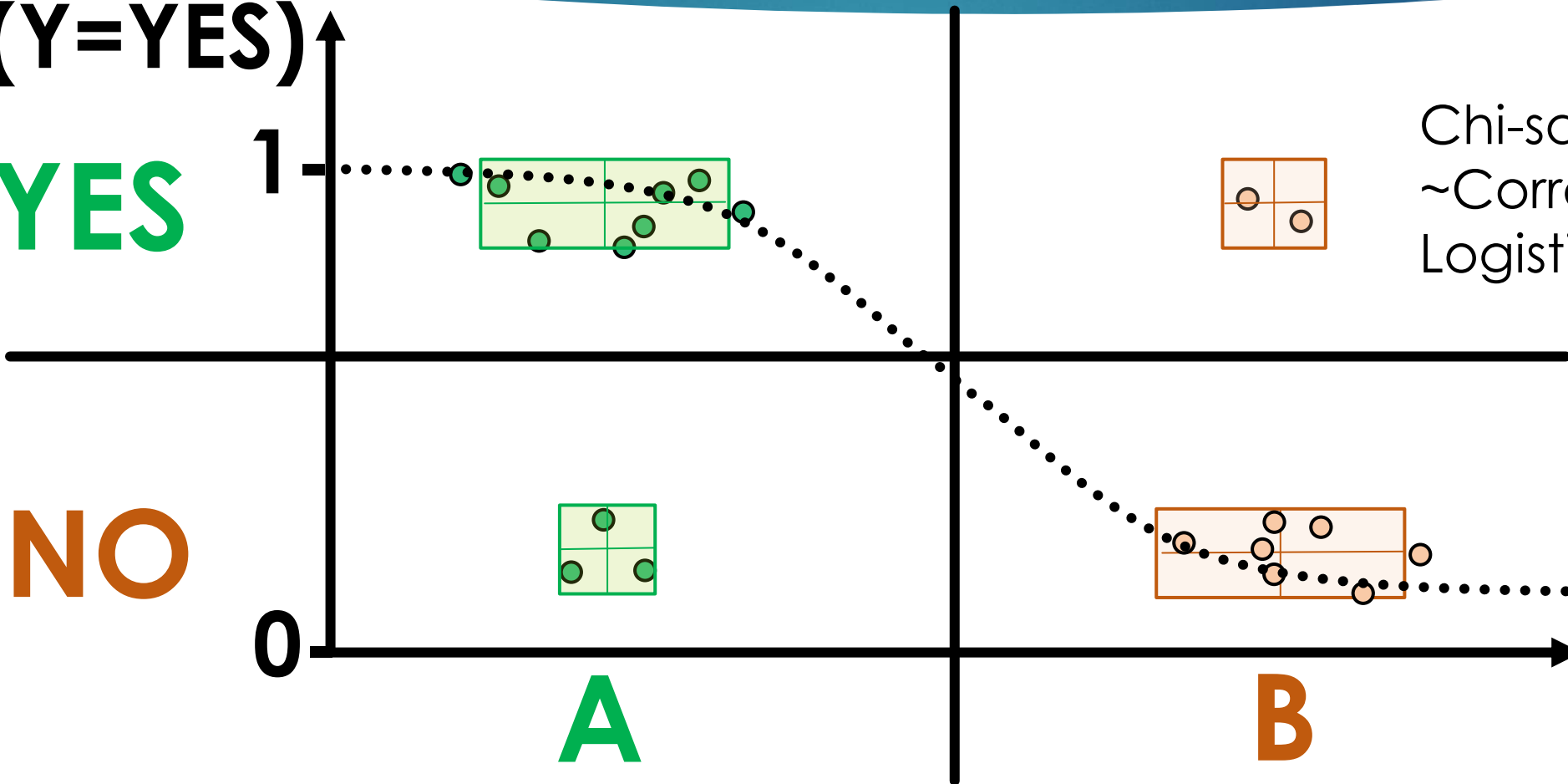
NO

0

A

B

Chi-square
~Correlation (r) [Phi Φ]
Logistic Regression



Central theme - Association

- ▶ Association between $X \leftrightarrow Y \leftrightarrow \text{other } X\text{'s}$
- ▶ 2 (or more) continuous variables (ratio/interval)
 - ▶ Normally distributed = Pearson's r
 - ▶ Non-normal, ordinal, limited intervals = Spearman's rho, Kendall's tau
- ▶ 1 discrete, 1 continuous
 - ▶ If 2 groups, t-test (or non-parametric Mann Whitney test)
 - ▶ 2 time points paired t-test (non-parametric Wilcoxon Rank Sum test)
 - ▶ If > 2 groups, ANOVA (or non-parametric Kruskal-Wallis test)
- ▶ 2 discrete
 - ▶ Chi-square (crosstabs)
 - ▶ (special case – small sample sizes), Fisher's Exact test
 - ▶ If paired = McNemar's test (and variants)

Models – horses of a different color

- ▶ Continuous outcome: Y output continuous
 - ▶ Linear regression (r^2): Y normal distribution
 - ▶ Poisson, Negative Binomial regression: Y count variable
 - ▶ Survival, Cox Proportional Hazards Regression: Y time to event
- ▶ Discrete outcome: Y output discrete
 - ▶ Logistic regression (χ^2): Y discrete (2 groups); X's continuous, ordinal or discrete
 - ▶ Other “flavors”: probit, ordinal, nominal regression
- ▶ Multiple Y's – compare groups, time points, multiple outcomes
 - ▶ Repeated Measures ANOVA (non-parametric Friedman's ANOVA)
 - ▶ Multivariate ANOVA



Correlation – Y, Covariates, X's (predictors)

	Y	cov1	cov2	x1	x2	x3
Y	1					
Cov1		1				
Cov2			1			
X1				1		
X2					1	
X3						1

Model of Y - Outcome

	Y	
Y	1	“Outcome”
Cov1		
Cov2		Covariates, typically adjusted for (e.g. demographics, risk factors) – assumed to be <u>independent of all other variables</u> and <u>measured without error</u>
X1		
X2		“Independent” predictors, group assignment, theoretically important – <u>assumed to be independent</u> and <u>measured without error</u>
X3		

Rest of Correlation - confounding

	Y	cov1	cov2	x1	x2	x3
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Y

Cov1

Cov2

X1

X2

X3

1	~0	~0	~0	~0
~0	1	~0	~0	~0
~0	~0	1	~0	~0
~0	~0	~0	1	~0
~0	~0	~0	~0	1

Covariates and "independent" variables assumed to be independent

Correlations assumed to be 0 or at least "small enough"

If not, then you have multicollinearity